

PRABHUKARNI

 +91 7013424340

 karniprabhu0@gmail.com

 [linkedin.com/in/karni-prabhu-aa7a68241](https://www.linkedin.com/in/karni-prabhu-aa7a68241)

 srikakulam, Andhrapradesh, 532185

Summary

Highly accomplished Site Reliability Engineer (SRE) with 9 years of experience driving operational excellence and reliability in complex cloud environments. Proven expertise in designing, implementing, and managing robust DevOps toolchains, infrastructure as code, CI/CD pipelines, and automated deployment strategies. Adept at leveraging cloud-native technologies (AWS, Azure, GCP) and observability solutions to ensure high availability, scalability, and security of critical systems. Passionate about continuous learning and applying cutting-edge SRE principles to solve challenging IT problems.

Experience

- VinylEquity**

SRE

- Automated AWS infrastructure provisioning and management across multi-account environments using Terraform and Terragrunt, ensuring consistency and repeatability.
 - Developed and optimized CI/CD pipelines with GitHub Actions and GitLab CI/CD, integrating Trivy and Jit for automated security scanning and vulnerability detection.
 - Implemented comprehensive AWS observability solutions including CloudWatch, Grafana, Prometheus, Loki, and ELK stack for proactive monitoring and performance analysis.
 - Deployed and managed scalable containerized applications on Amazon EKS, Elastic Beanstalk, and other platforms using Docker, Helm, and ArgoCD for efficient lifecycle management.
 - Configured External Secrets Operator (ESO) and Reloader to enable dynamic synchronization of AWS Secrets Manager with EKS workloads, enhancing security and reducing manual overhead.
 - Built and deployed serverless applications leveraging AWS API Gateway, Lambda, S3, CloudFront, and Route 53, optimizing for scalability and cost-efficiency.
 - Integrated AWS security services such as GuardDuty, Inspector, Macie, and Security Hub for centralized threat

Jan 2023 - Jul 2026

Skills

- AWS
- Azure
- GCP
- DigitalOcean
- Docker
- Kubernetes (EKS)
- AKS
- GKE)
- Docker Compose
- Docker Swarm
- Helm
- ArgoCD
- Jenkins
- GitHub Actions
- GitLab CI/CD
- Bamboo
- GitHub
- GitLab
- Bitbucket
- Prometheus
- Grafana
- Loki
- ELK Stack
- CloudWatch
- Headlamp
- Python
- Shell Scripting
- Go (Golang)
- JavaScript
- TypeScript
- Node.js
- WAF
- Cloudflare

detection and compliance monitoring.

- Established secure, keyless deployments by integrating GitHub Actions with AWS via OIDC.
- Centralized AWS access management by integrating AWS IAM Identity Center with Google Workspace and Cloudflare Zero Trust.
- Automated security monitoring and findings aggregation by integrating Jit with AWS and GitHub.
- Implemented Sophos malware scanning for S3 with real-time security alerts delivered via Slack.
- Secured API Gateway, ALB, AppSync, and Cognito using AWS WAF to protect against common web exploits.
- Deployed Lambda functions across sandbox, development, staging, and production environments using Terraform and Terragrunt for consistent release management.
- Proactively monitored and managed AWS costs by implementing billing alerts with Slack notifications.
- Configured OpenVPN and AWS Site-to-Site VPN to ensure secure access to private infrastructure and client networks.
- Integrated private backend services using API Gateway HTTP APIs and VPC Links for seamless connectivity.
- Deployed static websites on S3 with CloudFront and Route 53, enabling DNSSEC for enhanced security.
- Configured Amazon SES with DKIM, SPF, and DMARC to ensure secure and authenticated email delivery.
- Enabled private AWS connectivity by configuring VPC endpoints for S3, DynamoDB, Secrets Manager, and SES.
- Streamlined AppSync GraphQL schema deployments using AWS Amplify pipelines.
- Led end-to-end migration of on-premises workloads to AWS, encompassing assessment, architecture design, migration execution, and production cutover.
- Modernized legacy workloads by re-platforming applications onto EC2, RDS, S3, and EKS, significantly improving scalability and reliability.
- Developed AI applications using Amazon Bedrock, LangChain, LangGraph, Streamlit, and Python, building sophisticated AI-powered solutions.
- Designed and implemented RAG pipelines for efficient document ingestion, chunking, embeddings, retrieval, and context-grounded LLM responses.
- Developed LangGraph multi-agent workflows with robust state management, tool calling, routing, and task orchestration capabilities.

- Trivy
- SonarQube
- OpenVPN
- Site-to-Site VPN
- Sophos
- AWS Lambda
- API Gateway
- S3
- CloudFront
- Route 53
- AWS Elastic Beanstalk
- PostgreSQL
- MongoDB
- Neo4j
- Amazon Bedrock
- LangChain
- LangGraph
- LangSmith
- RAG Pipelines
- Streamlit
- Vertex AI
- Microsoft Foundry
- Model Context Protocol (MCP)
- ESO (External Secret Operator)
- Nginx
- VPC
- Cloudflare Zero Trust
- Jira
- Slack
- Terraform
- Terragrunt
- Ansible
- Puppet
- Chef

Language

- English
- Telugu
- Hindi

- Built and deployed Model Context Protocol (MCP) servers to expose application and infrastructure capabilities as AI-agent tools.
- Integrated Amazon Bedrock with LangChain and LangGraph to power LLM-driven applications and agent workflows.
- Leveraged LangSmith for comprehensive LLM tracing, debugging, evaluation, and observability.
- Developed interactive Streamlit interfaces for AI/RAG applications and enterprise knowledge retrieval systems.

• Techverito

Jul 2022 – Jul 2026

SRE

- Designed and deployed automated infrastructure using Terraform and Terragrunt across AWS environments, ensuring consistent and repeatable provisioning for high availability.
- Built and optimized GitLab CI/CD pipelines with integrated Trivy security scanning, enabling automated build, validation, and deployment workflows for faster release cycles.
- Deployed and managed applications on on-premises RKE2 Kubernetes clusters using ArgoCD and Helm, implementing GitOps-based application delivery and lifecycle management for improved control and visibility.
- Deployed and managed AWS EKS workloads through GitLab CI/CD, utilizing Helm and Kubernetes best practices to ensure scalable and highly available application deployments.
- Deployed and managed Small Language Models (SLMs) on Kubernetes clusters for AI-powered applications, integrating them seamlessly with existing application workloads.
- Designed and implemented RAG pipelines for AI applications, covering the full lifecycle from document ingestion and processing to embeddings, retrieval, and context-grounded LLM responses.
- Implemented comprehensive Kubernetes and application observability using Grafana, Prometheus, Loki, and CloudWatch, providing real-time monitoring across on-premises and AWS environments.
- Utilized Headlamp for efficient Kubernetes cluster management, workload monitoring, troubleshooting, and operational visibility across RKE2 and EKS environments.
- Implemented robust AWS infrastructure security using IAM, WAF, Secrets Manager, KMS, AWS Config, GuardDuty, Inspector, Macie, and Security Hub.

- Configured External Secrets Operator (ESO) and Reloader to securely synchronize AWS Secrets Manager secrets with Kubernetes workloads and ensure automatic refresh of application configurations.
- Developed Python automation scripts for infrastructure operations, deployment workflows, Kubernetes administration, and general DevOps automation, improving efficiency.
- Led on-premises to AWS migration and modernization initiatives, re-platforming workloads onto EC2, RDS, S3, EKS, and serverless AWS services to enhance performance and scalability.

• **Social Alpha**

Feb 2021 – Jul 2022

Devops engineer

India

- Architected cross-cloud infrastructure solutions across AWS, Google Cloud (GCP), and DigitalOcean, developing automated provisioning platforms with multi-version compatibility for Tomcat, Node.js, Java, and Python applications.
- Consulted enterprise clients on DevOps and cloud modernization strategies, including Terraform-based provisioning, blue-green deployments, and zero-downtime implementation across AWS and GCP.
- Designed and deployed serverless applications using AWS Lambda, S3, API Gateway, and equivalent GCP managed services, optimizing for scalability, cost, and performance.
- Directed multi-cloud DevOps teams, driving platform architecture, reusable automation, and infrastructure delivery pipelines using Jenkins and GitHub Actions.
- Implemented DevSecOps frameworks by integrating GuardDuty, Inspector, Macie, Security Hub, and SonarQube for automated compliance checks across cloud environments.
- Migrated Bamboo CI/CD pipelines to GitHub Actions and modernized deployment workflows across AWS and GCP environments, enhancing CI/CD efficiency.
- Configured Grafana, Prometheus, and Loki for centralized infrastructure and application monitoring across multi-cloud environments, providing unified visibility.
- Migrated on-premises Bitbucket and Jira platforms to cloud-based solutions and successfully migrated projects from Bitbucket to GitHub.
- Implemented cloud security and networking best practices using IAM, VPC, Nginx, SPF, DKIM, DMARC, and ClamAV for secure infrastructure and application operations.

- Developed Shell scripts to automate infrastructure provisioning, operational tasks, deployments, and routine DevOps activities, improving operational efficiency.
- Successfully migrated 6 complete projects from DigitalOcean to AWS, redesigning networking, IAM, storage, CI/CD, monitoring, and security to align with AWS best practices.
- Performed end-to-end cloud migration planning, execution, and production cutover for 6 applications, ensuring minimal downtime, data integrity, and seamless integration with AWS managed services.
- Leveraged Google Cloud services including Compute Engine, Cloud Storage, VPC, IAM, Cloud Monitoring, and Cloud Logging for cloud infrastructure provisioning, application deployment, and operational monitoring.
- Integrated

• **Integrated Data Systems Limited**

May 2017 - Dec 2020

Devops engineer

- Initiated career as a Software Engineer, gaining foundational experience in diverse technologies while transitioning through Linux Administrator and Automation Engineer roles.
- Automated daily scheduled tasks, user management, and disk management using advanced Shell scripting, significantly improving system performance and reliability.
- Managed comprehensive backup and recovery processes for systems and applications, and performed application patching on test servers to maintain system integrity and security.
- Oversaw service management, zone creation, and NFS mount setup, ensuring smooth and uninterrupted operations for critical services.
- Set up EC2 instances and configured all necessary prerequisites for seamless application deployments.
- Designed and deployed robust AWS solutions utilizing EC2, S3, Elastic Load Balancer (ELB), and Auto Scaling Groups for high availability and scalability.
- Provisioned AWS infrastructure using Terraform and developed reusable Terraform modules to automate cloud resource creation, enhancing efficiency.
- Created and managed multiple VPCs, public and private subnets, and established inter-region VPC connectivity using AWS VPC Peering for secure network segmentation.
- Installed and configured Jenkins, including essential plugins, security settings, and master-slave architecture to support scalable parallel builds.

- Designed and documented CI/CD tool configurations for streamlined automation workflows, promoting best practices.
- Managed infrastructure configurations using Ansible and implemented environment changes efficiently using Ansible Playbooks.
- Automated build deployments across various environments using Jenkins, ensuring consistent and reliable releases.
- Installed and configured Jenkins to support Java-based builds and integrated Jenkins plugins to automate continuous builds and publish Docker images to Nexus.
- Worked extensively with Docker, including building images, creating containers, writing Dockerfiles, tagging images, and pushing them to Docker Hub for efficient container management.
- Gained experience with Microsoft Azure, including deploying virtual machines, managing networking, configuring storage, and automating provisioning using ARM templates and Terraform.
- Experienced in Azure Kubernetes Service (AKS), covering cluster provisioning, scaling, workload deployments, and integrating AKS with CI/CD pipelines for automated application delivery.
- Worked with Google Cloud Platform (GCP), including provisioning compute instances, managing VPCs, and automating infrastructure using Terraform and Cloud SDK for efficient resource management.
- Experienced in Google Kubernetes Engine (GKE) for deploying containerized workloads, managing clusters, implementing autoscaling, and integrating GKE with CI/CD pipelines for streamlined operations.